the SPONGE

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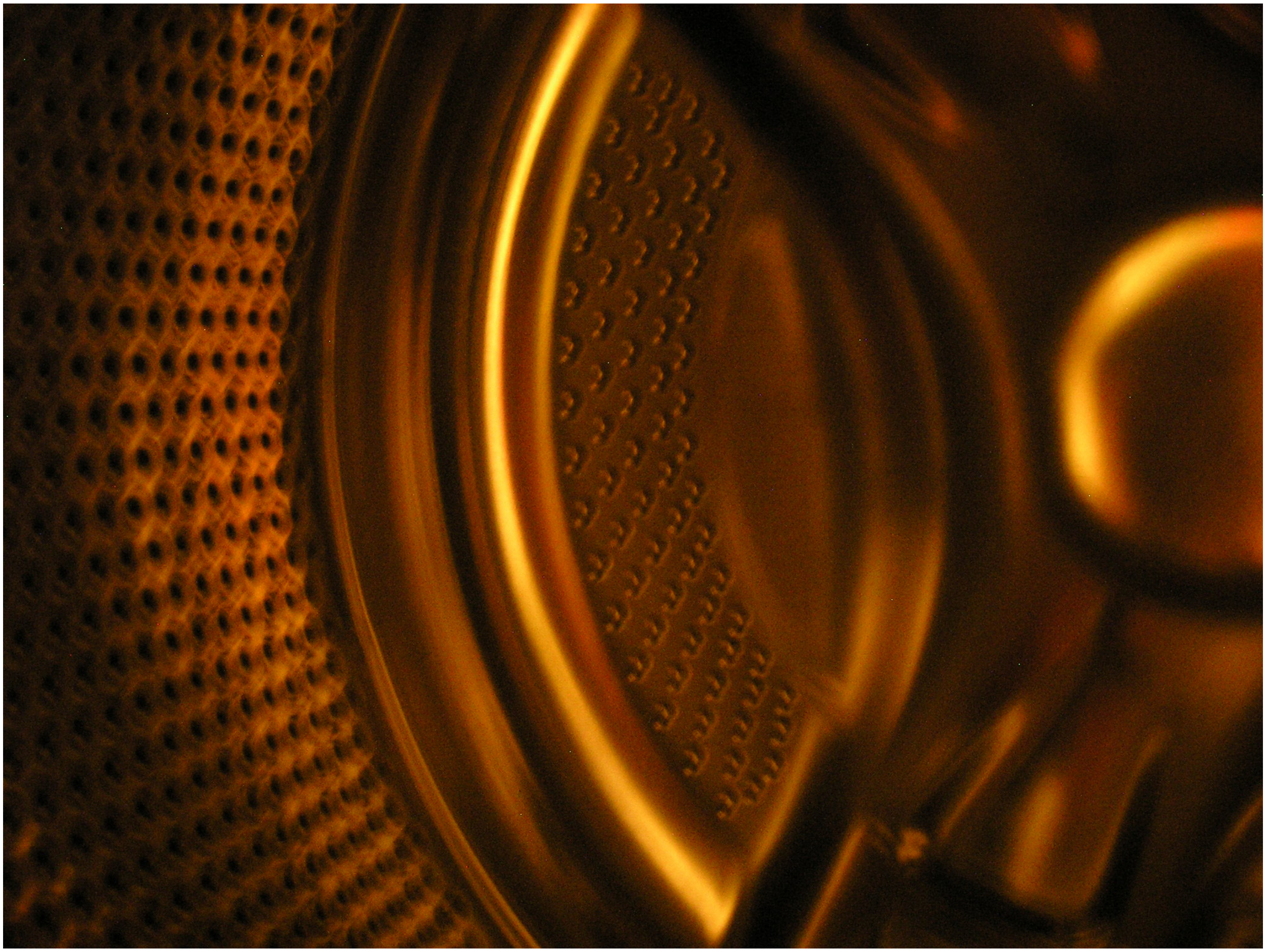

Photo contest winner: **Gravitational Domesticity** by Ariel Herbert Voss, Physics and Applied Mathematics

Additional content at: http://thesponge.eng.utah.edu/

The Sponge was founded in Fall 2010 to encourage and showcase the creative interests of the science and engineering community at the University of Utah. We publish anything from poems to scientific reviews, short stories to opinion articles, photography to cartoons, as well as ads for student organizations and events and practical articles helpful to students, once each semester.

We hope you enjoy this Fall 2011 issue and consider contributing to our next one! You can visit our website at http://thesponge.eng.utah.edu/.

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Sport Sociology Analysis of Rafael Nadal

Cheyenne Schmid, Staff Writer, Exercise and Sport Science

During the 2011 U.S. Open, professional tennis player Rafael Nadal endured a strange cramping incident in the middle of an interview after his third round match. His injuries from that match were also very noticeable. He lightly brushed this cramping occurrence off and told reporters he was fine and that it was just bad luck that it happened in so public a place. He clearly had experienced so many more injuries than what the public knew. As a reporter for ESPN stated, "Nadal appeared to be in intense pain and was incapacitated for nearly ten minutes…it was a window into the physical hardships of playing professional tennis."

Fans have been taken by surprise by the seemingly diminishing physical health of arguably the number one male tennis player in the world. People have seen him as invincible, just as spectators view the most elite professional athletes in every sport. Although Nadal just turned twenty-five (extremely young in society), he is really approaching old age in tennis terms. He began playing tennis at the age of four, which means he has roughly been playing for over eighty percent of his lifetime. Nadal told reporters, "I think for everybody is tough. Maybe for me is tougher because I started earlier than everyone else. Nine years in tennis -- that is a long time. Many people do not play that long." It seems as though retirement is finally coming to light as the fans realize that both Nadal's physical style of play and his tough schedule are incredibly taxing on his body.

Rafael Nadal's injury and physically devastating playing history could easily be seen as deviant overconformity. According to sociologist Jay Coakley, deviant overconformity consists of supernormal ideas, traits, and actions that indicate an uncritical acceptance of norms and a failure to recognize any limits to following norms (playing through injury and/or painkillers/drugs) and can be a very significant problem in sports. Nadal even admits to going beyond the limits and says, "when you put your body at the limit, you're going to have problems."

As researchers have shown, both elite athletes and coaches use a sport ethic to assess actions and mind-sets in the social world of competitive sports. This sport ethic, according to Coakley, includes four norms: athletes are dedicated to the games above all other things, athletes strive for distinction, athletes accept risks and play through pain, and athletes accept no obstacles in the pursuit of success in sports. Rafael Nadal clearly qualifies for those four distinctions. He puts all his time into both actual tennis and the physical training for it, has strived to become number one and continues to fight to uphold that title, plays through pain/injuries while knowing the extreme damage his body is enduring, and is not willing to ease up on his playing by sacrificing physical health in order to become the best.

Rafael Nadal is a prime example of deviant overconformity as he sacrifices, dedicates, and willingly puts his body in danger for the sake of tennis.

International Blasphemy Rights Day 2011

Jason Cooperrider, Staff Writer, Neuroscience

On September 30, many around the world celebrated the second annual International Blasphemy Rights Day (IBRD). IBRD was created in order to commemorate the anniversary of the publishing of the Muhammad cartoons that sparked global outcry (including some violent protests) by Muslims who believe it is blasphemy to draw an image of Muhammad, the divine prophet of Islam. In commemorating this anniversary, the purpose of IBRD is to promote freedom of speech and expression without giving special privileges to religion or any other institution.

IBRD, also known simply as Blasphemy Day, was started by the Center for Inquiry, which has the following to say about it:

> The purpose of this event is to set a particular day as a day to support free speech, support the right to criticize and satirize religion, and to oppose any resolutions or laws, binding or otherwise, that discourage or inhibit free speech of any kind. The focus on 'blasphemy' is simply because it is such a salient issue, and one for which a lot of consciousness-raising is necessary.
>
> The goal is not to promote hate or violence. While many perceive blasphemy as insulting and offensive, it isn't about getting enjoyment out of ridiculing and insulting others. The day was created as a reaction against those who would seek to take away the right to satirize and criticize a particular set of beliefs given a privileged status over other beliefs. Criticism and dissent towards opposing views is the only way in which any nation with any modicum of freedom can exist.

Blasphemy
If you're going to do it, do it with style

Former United Nations representative Dr. Austin Dacey, in his book The Secular Conscience and in a talk he gave at the University of Utah in February 2010, suggests that religious ideas/beliefs should not be exempt from critical inquiry or public discussion any more than any other idea/belief should be. This holds even for what certain religions might consider blasphemous. To make such ideas/beliefs exempt would fly in the face of freedom of speech and expression, a critical aspect of a true democracy. Dr. Dacey says that it is people who deserve inherent respect and should be treated accordingly, not ideas or beliefs.

To honor IBRD, I composed the folllowing limericks. They appear below, along with a Demotivator-style picture that I think represents IBRD well.

There once was a being called God,
whose actions seemed rather odd.
He created big floods
to wipe out His duds
and expected the survivors to applaud.

There is a method called science;
on evidence is its reliance.
It preaches reason,
which, to faith, is treason.
And it confronts religion with defiance.

Darwin posed the theory of evolution,
which comes to a heretical conclusion:
life evolved from scum,
not God's aplomb.
For religion, this causes confusion.

"This is blasphemous to Allah!"
shouted the Muslim from his sauna,
when he saw the cartoons
of the Prophet lampooned.
It was now time for a fatwa.

Hinduism contains many gods,
who sometimes act like sods.
And, according to Hindi belief,
cows should never be used for beef,
so vegetarians get friendly nods.

Isolated Incident

Alex Eldredge, Bioengineering

(Photo available online)

Never mind, they sleep
For the longest time they projected stars upon
their mind
Seeing eternity upon their fingers, he wept
Together into the longest night, funded by a multi
cellular opera they sleep

Foraging for you self
You pass on through
Black out into the light
Unto the consciousness that has no feelings
They sleep

50 miles into the desert
Feeding upon cacti and lizards
Polar bears wander into the city streets
Forgive, forget, and kill
As the water turns into liquid bromine
His footsteps turn into fire, brought on by vodka
flavored settlement's sediment
As he looks back the scene changes into an unfa
miliar colloquialism
He finds little concealment in the scenery

He walks back down
Into the crowd
As he is talking to the many quarks
He owns the f***ing world

Foraging for you self
You pass on through
Black out into the light
Unto the consciousness that has no flight
They sleep

He is lost inside in the middle of his unconscious
Forgetting reality with every step
He laughs as the buzzards sing a suite
They follow the path down the cliff
The fishes harness the wheel
Driving him down, all the way down
He wakes into a daze

2:30 Am, afraid to fall back asleep
He slowly and numbingly slips back into his mind
A new thought, he is intrinsic to this new and
strange land
He never finds the glass, an unkempt biosphere
that will linger on with him
He looks back at her and sleeps

Winter Joy

Hans Knecht, Staff Writer, Chemical Engineering

Compared to Persephone, the falling snow
Just as appealing and mourned in passing
Yet has more power
Than Olympus combined

Gods know no power like that of snow;
It puts some to sleep, and keeps some awake.
The power of whiteness
Hope a burden of lightness

Some awake, some asleep
Gaea has laid us low
Humans so mighty in moderation
How we weep, when Divine announces

Why, why do we find depression in snow?
Why no rejoicing? It's baptismal water.
Is not this a chance for ascension;
A new chance to prove our worth?

We reached a fork,
We chose a path,
We strive for perfection
And choose again the path less traveled.

Mirrors

Nicholas Roehner, Bioengineering

He asks me,
For whom is your strength?
It is for myself, and no other.
For whom is your love?
It is for no one and for all.
Proud I am.

She asks me,
For whom is your love?
It is for no other, and myself.
For whom is your strength?
It is for all and for no one.
Humble I am.

Liar, they say.
How can you humble yourself with such pride?
Because I am not one.
I am two.
My mind reflects thee.

And if thou should die,
Know that this contradiction lives on
Inside me and that I
Take pride in such humility.

Entropy is Not Our Anomie

Sam Leventhal, Physics/Applied Mathematics

Intent:

After building a proper backdrop I hope to show that entropy has dissolved ideal forms, forced us into action, and most importantly made us creators. By not fighting our driving forces, by not acting as the bow against the flood which is soon to break but rather bending in its direction, we can take rein and use the force of our environment to our advantage. We can escape apathy and bear the burden of our changes and of our effects, allowing us to live as an aesthetic who takes art as life, and purpose as life's effect.

Background:

Neutrality

Religion and other forms of blind faith are evaporating. Beliefs that once filled all minds in a frighteningly beautiful harmony, like that of marching boots or monotone chants, have begun to lose value. This loss of old religion has been the natural effect of the sciences and a growing knowledge; we have dissected faith and found only simple mechanics. Born are the mentalities of nihilism and existentialism, which are built upon the non-existence of truths outside of human conception. For nihilism moral constructs are a will to power and a simple solution to the unknown. The Judeo-Christian moral order and good/bad dichotomy was born as a "slave revolt in morality," where the resentment of the lower class towards aristocrats, or subject towards master, is comforted by a religious scheme. Where despite class pure souls hold eternal bliss and the sinister, regardless of power, are condemned (Nietzsche, On the Genealogy of Morality). Less ascribed to the philosophy of will to power, existentialism holds to the belief that existence precedes essence, that during our existence we build a valuation (Sartre). What is overlooked with these philosophies is the new faith they provide. A new harmony of different shape and color. To avoid the blanket of apathy that now seems too common during this transition from god given purpose to a complete responsibility of actions the importance of humanism, and beauty of our godless circumstance must take religion's place. It is a weightless freedom to recognize we have not been resting against the given purpose ascribed to us by religion but rather have been clinging to, preventing each individual from rising.

|S|

In an isolated system (the Universe being an example) entropy, change, and spontaneity will increase. For thermodynamics this is seen as the increase of thermal energy. The inevitable increase of entropy for any closed systems has come to be known as Heat Death. Statistical entropy is defined in terms of the micro-states composing a macro-state. The micro-state is the microscopic configuration of the total macro system. The increase of entropy from this perspective means an increase in micro-states. A closed system growing in entropy therefore grows in micro-states, in constituent variation. In nature, this is seen as a loss of organization, movement towards the chaotic, and a lessening of order with time. Take for example a puzzle: the micro-states are each piece and whether or not they're connected to any other piece, the macro-state is the total configuration. Zero entropy would be the completed image, but a puzzle will not spontaneously build itself, work is required. Entropy's natural progression towards change is the cause of our understanding of time. Time is the physical culmination of entropy, where time's arrow is in the direction of entropic increase.

Being in Time

To be self aware is the ability to relate the self as belonging to multiple events, to exist as a single persona, as a constant state, through different times. The changing time implies a change of states and situations. The spanning self is a projection of consistency within the growing entropy of our environments. We paint the self as a solid image, a completed puzzle, which travels alongside a universe in constant change. This tendency becomes our identity. The ego is the house left standing during a war. This self-definition as a solidarity having existed within past and current environments is the first aspect in the construction of an ego. The second defining feature comes from the mind's understanding of inevitable entropic progression, change, and an arrowed time. This faith in change forces the self to question future states, and as a result how we may affect these states. We are forced to act as an entropic affect, making man as affect the second aspect of ego. We are forced not only to exist in constant change, but we must also contribute to that change. To do nothing is impossible, entropy will increase, and our hand, be it still, is an affecting force. How then we dissipate entropy becomes a feature of the ego. Are you to build pyramids bound to crumple or bombs to hasten the process? But what is most unique in humanity, is a common desire to decrease entropy. A desire to delay decay through organization, construction, and art. To summarize: the ego is an illusionary projection of solidarity within a state of decay which recognizes that, subject to change, we hold chaos within ourselves, and it is chaos which reshapes environments.

Primum Movens?

Is man able to choose in which direction to propagate entropy? Is man free? We are not the originating cause any more than a rock is, which when dropped into a pond produces a wave. With respect to only the rock and pond, the rock is the protagonist, but by expanding the system the cause which dropped the rock becomes the originator. By continually expanding perspectives in this way we find additional sources. Say there is a tree that fell on a mountain that freed a rock to roll into a pond to produce a wave. The tree becomes the causing force. Can this process of expansion continue infinitely? Or could there be a single, predictable event, say a calculable big bang? In either case, man is still no originating cause. Human action is the response of environment and history. Our actions are biologically and genetically induced, and our actions shape our environments which we later choose from. Like the wave source, we are the culminating cause of other effects. We are born as the effect of past events. The effect of other recursive causes. Man is no cause, but rather has cause flow through him.

Relatively Free

Unlike the rock/pond scenario, however, our system is more complex, composed of countably infinite affecting agents. The complexity of our situation would be like dropping two rocks, each producing waves of constructive and destructive interference. Or, more realistically, when dropping the rock it is raining, and with each droplet of rain comes a new ripple to interfere. Each effect on our timeline is a single thread. Take the numerable threads and braid a single rope: the rope becomes the individual. The complexity of the human situation is the reason why man has until recently been seen as a free agent. However, our freedom is an accurate approximation. Because of the near-infinite causes, the complex intertwining of many events, we are able to continue viewing ourselves as free. Similar to the birth of Einstein's relativity, Newtonian mechanics no longer could be seen as a natural law, but an accurate approximation at low speeds. Where low speed includes any speed you will encounter during your life. Our freedom, though non-existent, is relatively true. The relative complexity of entropic growth with respect to our ability to understand cause and effect allows man to be approximated as free-willed. Man can be approximated as a source despite being the intersection of various paths of spontaneous change.

Implications:

Forfeit of will, our single freedom

One freedom does exist in itself and not as an approximation. To be free we must escape the drives and causes that induce our "choices." Even though all actions are produced by systems of cause and effect, an exception exists: randomization. Random can have no originating cause. It is an event void of precursor, and comes into existence in itself. Our freedom then is to give up our choice and act at complete random. To make a choice, for example, and to have that choice be random, you could number your options then use a random number generator to determine which one you will act on. For example when confronted with a choice you have even numbers mean yes, odd mean no, or: if odd take the path right of the mountain, even the path left, and if prime go over the mountain. Though we are the hand of cause and effect, we are able to choose to act at random and escape our innate driving forces. Choosing to act random is our free choice, though its outcome is uncontrollable by us. A seeming paradox arises: our single freedom, allowing us to escape our mechanization of having our actions/choices be the effect of other causes, relies in our ability to choose to forfeit our will and act independent from other causes, meaning, to act randomly. To be free we must choose to make no choices.

Unnamable

All forms exist in a continually changing state due to entropy (time) and are constantly reshaping, gaining new attributes, or becoming new objects. Assigning names becomes impossible. To identify an object in few words requires talking of an object as it was, ignoring how it has changed and what it has become. A temporary solution is to redefine what the name represents. An example is the ego. We hold an identity to ourselves throughout life despite our innumerable changes in personality and beliefs. I have a consistent idea of my ego by considering my younger self to be the same person as I am now. However it would be impossible to have an accurate description of my identity if I had stopped defining my identity at age twelve. The ego is an example of a constantly changing form always described by the same name. The "it" or "I" must undergo entropic change after being named and this image we impose when naming does not last, but fades, and looses solidarity. With this comes a loss of unitization and a falsity in titles. Names are then only imaginative lingual tools, generalities and not actualities. There does not exist a distinct divide between an object and its environment; all objects evolve with their environments. Objects are all changing, graying the separation between "it" and all else. Universal entropy illuminates the fiction of language and identities. In truth, no item can be generalized, but rather each item at a very specific time is unique and will immediately change. No general form of an item exists and no single unit can remain constant to its self. Entropy is corrosive to language and things become unstuck to their names. As a result the atomistic projection of language cannot be taken as truth but as a beneficial tool. Names serve to generalize and identify common attributes between objects, but the objects themselves are directly tied to time, since their entropic state must change, and their environment, since during evolution they interact with the external world. The entire system is then not composed of units such as man, earth, star, and galaxy, but as a totality, a continually reshaping form in time where all objects we name are directly tied and inseparable from their surroundings. The universe without language (a separatist atomization) takes the form of an All State which is a single continuous and reshaping form containing all things which are naturally inseparable.

Man's Reaction

Progression and alteration is forced upon us. If we are to remain conscious, we will always be subject to a changing environment and a changing self. Two forms of man come from this entropic state. One sees only disparity and apathy, the other opportunity. The former are overwhelmed by the burden of change. The despondent see their situation as an aimless descent. They are rag dolled by the elements, where chaotic winds constantly change their path, and they land only where exterior forces have brought them. Unable to take rein, the apathetic allow the universe to improvise in their place. The latter of the two acknowledges their condition, accept themselves as another means for entropy to propagate, and as a result are able to choose the focal points to direct this change. Pregnant with the future, the opportunistic look on our entropic circumstance as a gift.

Thou Art God

Man is forced to act as change when viewed as a self-defining node through which entropy transgresses. To be a cause for change is to create. Some are overburdened by this responsibility and use destruction as their only form of creation. These are the apathetic and despondent. Others develop the changing self, focusing their wave fronts. The self affirming individuals understand themselves to be a creator, granting the perspective of life as art. Is not art one's own creationism? Man's identity is then an aesthetic projection, for our existence is an act of perpetual artistry. In truth little choice is involved. Our creationism, our art, is forced. Our art is the effect of culminating threads. We then form a fictional projection of self, wound from these threads, and this self is inevitably an aesthetic phenomenon. More practically however, being relatively free willed, we may use our freedom to build the aesthetic individual. The despondent

forget life-purpose out of spite, feeling wronged that value and morality is not the third element after space and time. They claim each day is their purpose. For only in the sun's passing can you jump over the earthly burden of your shadow. The artists know the shadow to be a template; a form which they can give character and color. The artists work to become these shadows, praising noon, the time when they are nearest to it. As creators we may create ourselves and master our artistry, where artistry becomes a synonym for life.

The Aesthetic Experience

Life as art means to understand that art and aesthetics serve to illuminate the human condition. The aesthetic experience comes without effort or consent. What's more, the spectator is exposed to an element, realistically neutral in existence (the movement of air at varied frequency, a drop of oil with dye, a urinal) and unavoidably experiences it in a very non-neutral manner. The aesthetic experience is as unavoidable as laughing when being tickled, and equally as forced. It comes as a will-less knowing. We cannot approach art looking to exhibit authority over it, rather, the temperament with which art appeals is one of receptivity. As aesthetic phenomena we are subjugated by art, and this subjugation is the aesthetic experience. The aesthetic experience acts as a window onto our aesthetic existence, a mirror reflecting the human condition. Forced into an aesthetic experience through art we are shown how creationism and artistry is the human response to existence.

On Evolution

Life as art, but to what ends? A short lived aesthetic insight or a simple satisfaction? But what matters my happiness? What matters my epiphany? My anger, or my tranquility? Is not all a neutral change? Our life is our responsibility. A burden to be taken as a light and playful jest or as a weight to drag each action. No longer is there "thou shalt" but rather: "and so I make it." Responsible for the value our actions hold, becoming apathetic is all too easy. Too feasible is it for man to become stagnant water. But remember, no man is an end in himself. To live for one's self is to act onto the environment that effects all. Remain aware your artistry uses existence as its canvas. What's more, the child of our changing universe is evolution. All change is inevitable. Our adaptation is inevitable. We change the environment that changes us allowing us to use our creationism, our artistry, to make man our masterpiece.

Neo-Bedouin

Elliot Smith, Neuroscience

I lean on a camel stick and stare
Into thick air,
Knowing god's gift is going to get me there.

My one humped habibi dune hurdles
Limping towards a well for one ladle
And a stagnant sludge gurgle

Arabian sands, date palm stands and
Strands of prayer beads serve my needs, as
Falcons that catch rabbits inhabit my dreams
While mirages gleam
Past hot coffee pot steam,

Tabla Drums babble A salaam w'alaykums
After dawn comes, and minarets scream.

They advise I rise, ablute and pray
As I wonder which SUV to take today
Two backed up blocks to an air-conditioned café
For political discussions and a
Gourmet luncheon. I pray

Again before visiting my sky-impaling
Office to pop the top off
Another capped oil crop,
As gas is going up, and I tend to spend a lot.

Night yields respite from sun's burning light
So we fill it with the neon type
And international flights
To Saudi, Levant, Europe, Tunisia
Soaring over my snack of shawarma and sheesha.

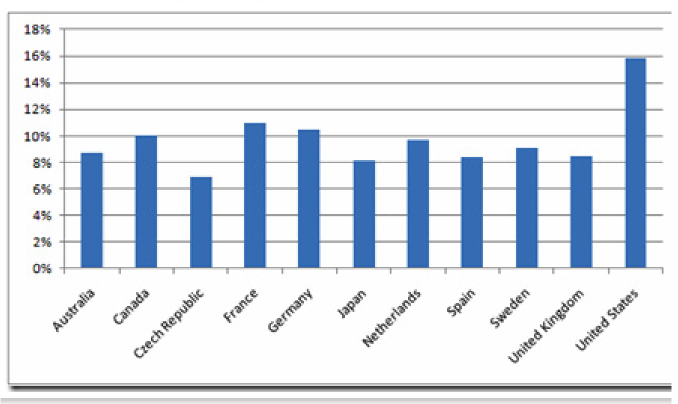
Photo by Elliot Smith, Neuroscience

The Economics of Malpractice

Dallin Hubbard, Staff Writer, Bioengineering

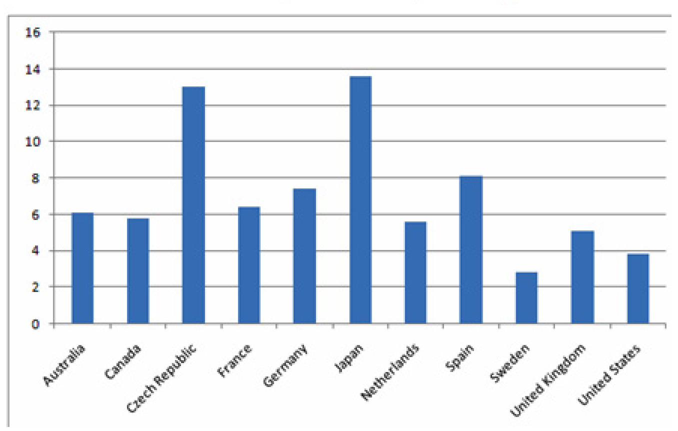
As noted in recent statistics, healthcare costs are very high in the United States as compared to other countries (see figure below). Some controversy has ensued around inflated malpractice lawsuits and ambulance-chasing lawyers seeking to benefit from the claims of injured (or not-so-injured) plaintiffs. As the government gets closer to legislation of caps on malpractice lawsuits, the debate has come to the forefront of rising healthcare costs.

Healthcare spending as a % of GDP: 2006



Source: Organisation for Economic Co-operation and Development

Doctor Consultations, Number per Capita: 2006



Healthcare expenses and doctor consultations by country [1].

While the arguments seem convincing to the layman the actual reasons for healthcare's rising cost and the actual contribution of litigation seem to be minimal. Only 2% of plaintiffs ever seek compensation for injury through lawsuits [2]. Also the volume of tort litigation (tort litigation is "private or civil law wrong or injury, other than breach of contract, for which the court will provide a remedy in the form of an action for damages," Black's Law Dictionary) has actually declined steadily since 1990 [3]. Bruce Finzen in the journal Minnesota Trial Lawyer actually states the cause of increased insurance premiums is the insurance companies themselves:

> Although medical malpractice insurance costs have been on the rise in the past few years, these increases have little to do with 'out of control' jury verdicts. Rather, the primary culprit is the insurance industry itself. Insurance rates reflect conditions in the financial marketplace. With a turbulent economy post 9/11, insurance-company investments have floundered, forcing the industry to increase rates to maintain profit levels [4].

The Wall Street Journal has pointed out that "the litigation statistics most insurers trumpet are incomplete," and the U.S. House of Representatives has called for an investigation of the insurance industry's responsibility in causing a nationwide malpractice crisis [5].

The New York Times reflected this idea of inflated litigation cost statistics. Tom Baker answers the question from the author Anne Underwood:

> Question: 'But critics of the current system say that 10 to 15 percent of medical costs are due to medical malpractice.' Answer: That's wildly exaggerated. According to the actuarial consulting firm Towers Perrin, medical malpractice tort costs were $30.4 billion in 2007, the last year for which data are available. We have a more than a $2 trillion health care system. That puts litigation costs and malpractice insurance at 1 to 1.5 percent of total medical costs. That's a rounding error. Liability isn't even the tail on the cost dog. It's the hair on the end of the tail [6].

While medical malpractice rates have maintained a steady level since epidemiological studies done in the 90's, the number of injuries has risen, actually decreasing the percentage of cost from malpractice lawsuits [6]. This means that during the past years the cost of malpractice litigation relative to total healthcare costs has gone down, undermining the explanations for rising health care cost stemming from litigation.

But still the anti-litigation sentiment has pushed through to the House of Representatives and to the President himself. Barack Obama was quoted as saying curbing frivolous lawsuits against doctors could help bring down medical costs [7]. So the honest question remains: how much of this malpractice litigation is really truly beneficial and how much is frivolous? Can a monetary limit be set on frivolous? That question is difficult to assess.

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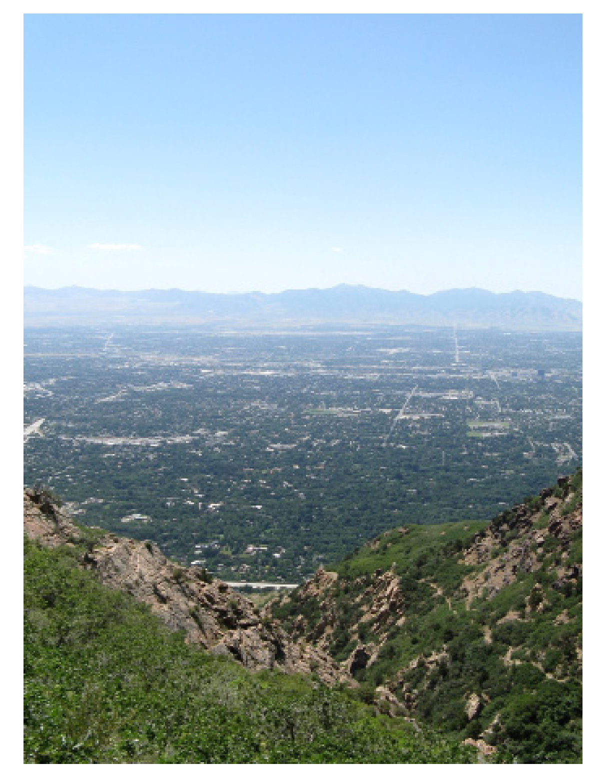
Outdoors

Mr. D, Bioengineering

With the Fall semester halfway over, and winter approaching, now is as great a time as any to get outside and explore Utah's great outdoors, where anyone can find something to do that they enjoy. If you have no clue were to start, no worries, this column is here to help you do just that. With your help, every issue we will discuss some outdoor activities, gear, and maybe even some pictures or videos (ok... so you are going to have to look at the online version to see the video). So if you just went on an awesome hike, or found an awesome bike trail, or have some crazy skiing pictures jumping out of a helicopter let us know at The Sponge and we will publish it here.

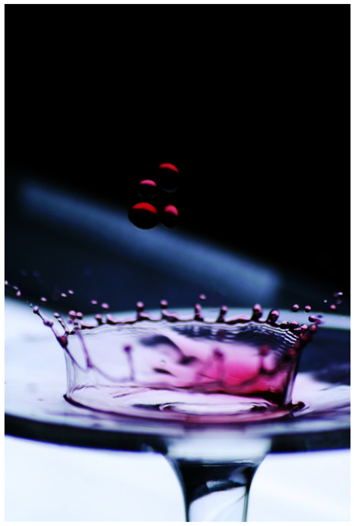
With the weather getting cooler, now is a great time to get some last minute hikes in without the risk of heat stroke. If you are looking for something close to Salt Lake City with a little bit of a challenge to it but some great rewarding views when you get to the top, you should check out Mount Olympus with a trailhead along Wasatch Blvd at 5800 South. Be warned though, it is a mostly Class 2 with the scramble to the peak being Class 3. If you are not sure what that means, search for Yosemite Decimal System. The trail is about 3.5 miles long in one direction and goes up over 4,000 feet. Depending on how fit you are, it will take anywhere from half a day to a whole day to do this hike, and you can even make an overnight camping trip out of it. Of course, if you decide to do this hike MAKE SURE YOU DO YOUR HOMEWORK!!!



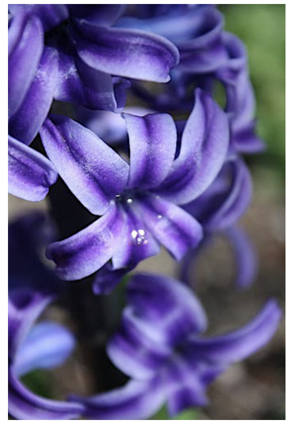
Looking west over Salt Lake Valley from about half way up Mount Olympus Trail. The Oquirrh Mountains can be seen in the distance with the highest elevation being Flat Top Mountain at 10,620 ft. July 2011.

Hiking, just like any outdoor activity, has risks. Make sure you are prepared for your surroundings (animals, weather, etc.), do your homework before you go, and let someone know where you are going and when to expect you back. If you plan on doing outdoor activities, an essential piece of gear worth investing in is a hydration pack and water bottle. Keeping hydrated is important, and you should carry at least 0.5 liters for every mile you hike.

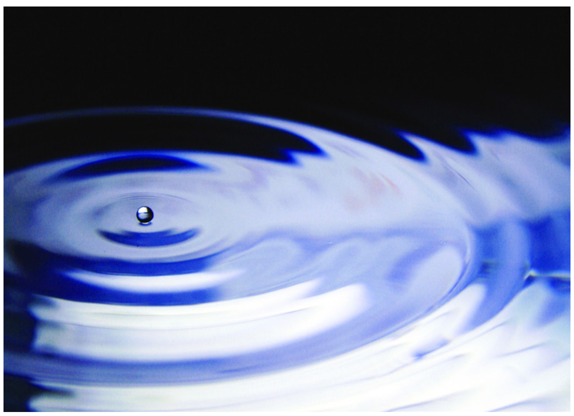
If there is something you want to see here, or want to submit some photos or a cross-country ski trail you just found, send us an email at uusponge@gmail.com and put "Outdoors" in the subject line.

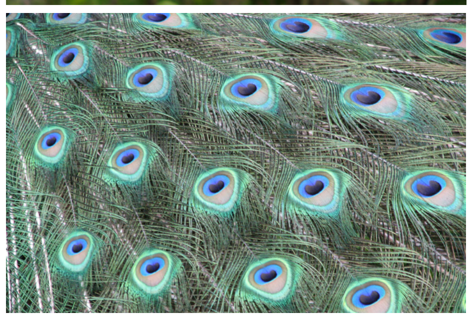


Flower (below) and *Garden* (right) by Judit Barabas, Biology



Rice Paddies (above left) and *Caterpillar and Frass* (below right) by Sarah Knutie, Biology
A Crown of Wine (above right) and *Wave* (below) by HwanJune Cho, Electrical Engineering





Untitled (above) and *Untitled* (left) by Lindsey Reader, Biology
Beargrass (below) by Johanna Varner, Biology

Cactuspic (above left) by anonymous
Untitled (above) and *Untitled* (below) by Peter Yihui Li, Bioengineering
Serving Passiflora (left) by Kipp van Schooten, Physics